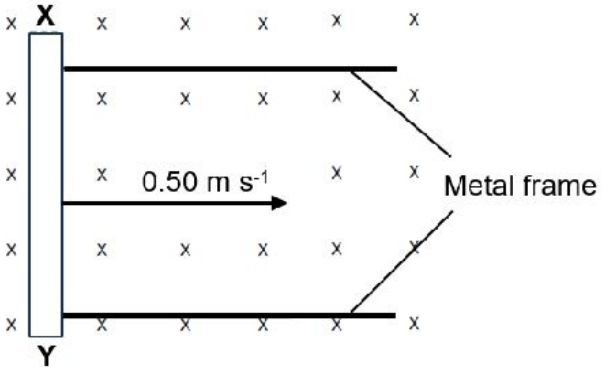
26	A straight metal rod XY of length 24 cm moves at a constant velocity of 0.50 m s^{-1} along a very long metal frame. The rod is moved in a region of uniform magnetic field of flux density 0.15 T directed into the plane of the frame throughout its entire length.  What is the magnitude of the magnetic flux cut by the rod in half a minute?

	A	0.018 Wb	**B**	0.036 Wb	**C**	0.54 Wb	**D**	1.1 Wb

L2 **Answer: C**

Flux cut, $\Delta\Phi$ = BA = BL(v x t)
Area covered in half minute = 0.24 x (0.50 x 30) = 3.6 m^2
Flux cut = BA = 0.15 x 3.6 = 0.54 Wb